REDUCING PACKAGING DEFECTS THROUGH STATISTICAL PROCESS CONTROL

Rice Packaging — Process Capability & SPC Analysis

Emmanuella Owa

Process Quality Engineer

> Identified a 0.56 g process-centering gap responsible for capability shortfall and developed a control strategy capable of improving Cpk from 1.24 to above customer requirements.

TOOLS & METHODS

Minitab · Capability Sixpack · Xbar-R Charts · Anderson-Darling Normality Test · Cp / Cpk / Pp / Ppk · DMAIC

OVERVIEW

Executive Summary

A rice packaging facility received customer complaints about inconsistent bag weights. This project applied DMAIC and Minitab's Capability Sixpack to 100 production measurements across 20 subgroups to diagnose the process, confirm statistical control, and quantify capability against a minimum Cpk of 1.33. A special-cause event was identified and removed, revealing a stable but off-centre process requiring targeted re-centering.

Business Problem

Industry	Food & Beverage — Consumer Rice Packaging
Product	Consumer rice bags, 750 g nominal fill weight
Specification	750.00 g ± 5.00 g (LSL = 745 g, USL = 755 g)
Requirement	Customer minimum: Cpk ≥ 1.33
Problem	Customer complaints of underfilled bags. Internal audit flagged potential capability gap. Risk of regulatory non-compliance and product returns.
Scope	Single production shift, one fill line, 100 measurements (20 subgroups, n=5)

Objectives

- Assess process stability using Xbar-R control charts; identify and investigate any out-of-control signals.
- Verify normality of weight distribution — required assumption for Cp/Cpk indices.
- Calculate and interpret Cp, Cpk, Pp, Ppk against the customer requirement.
- Provide quantified, data-driven recommendations for achieving and sustaining Cpk ≥ 1.33.

APPROACH

Methodology — DMAIC Framework

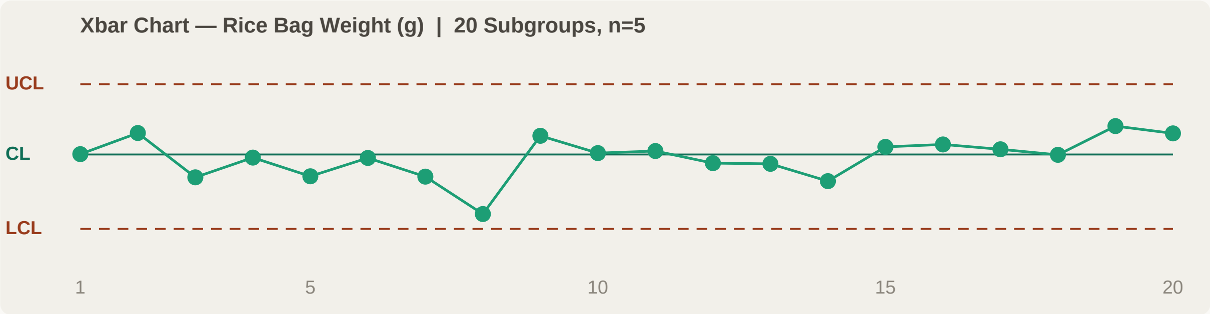
Phase	Activity	Tool / Output
Define	Scoped problem statement; defined CTQ characteristic (bag weight) Voice of Customer: Cpk ≥ 1.33	Project Charter VOC Analysis
Measure	Collected 100 bag weights; 20 subgroups of n=5 Sampled every 15 min over one full production shift Verified measurement system (calibrated scale) before study	Data Collection Sheet Calibrated Weigh Scale
Analyze	Xbar-R chart: confirmed stability; identified OC point at Subgroup 8 Root cause investigated; special-cause event confirmed (equipment adjustment) Subgroup 8 removed; re-ran capability on stable dataset Anderson-Darling test confirmed normality (p=0.838) Capability Sixpack: Cp, Cpk, Pp, Ppk calculated	Minitab 21 Capability Sixpack ANOVA
Improve	Identified process mean shift as root cause of Cpk shortfall Quantified mean adjustment needed: +0.56 g to achieve Cpk ≥ 1.33 Developed fill-head adjustment protocol	Process Centering Plan
Control	Proposed Xbar-R chart for ongoing production monitoring Defined reaction plan for OC signals Recommended fill-head set-point procedure update	Control Plan SOP Update

RESULTS

Analysis & Key Findings

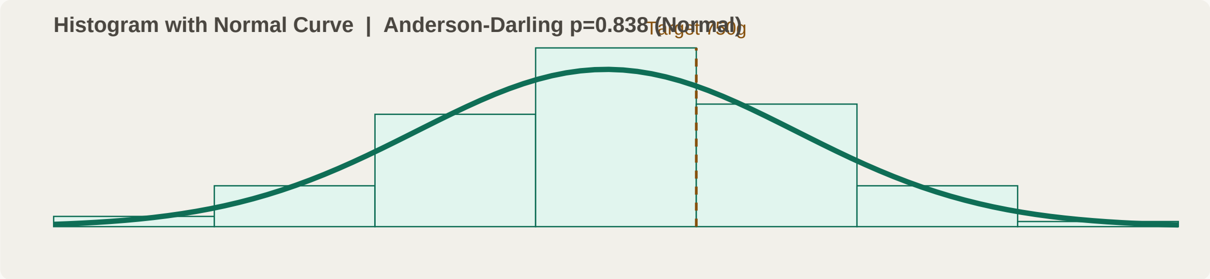
1.41	1.24	1.45	1.28
Cp — Potential	Cpk — Actual	Pp — Overall	Ppk — Overall Actual

Xbar-R Control Chart — Stability Assessment



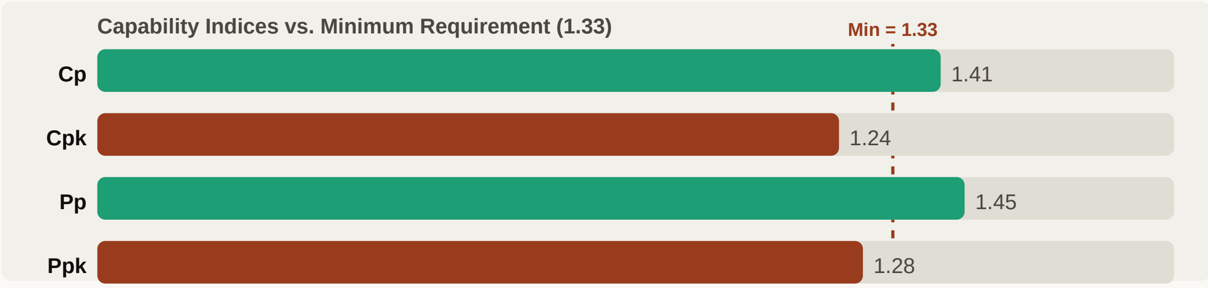
Subgroup 8 (mean = 747.72 g) fell below the lower control limit (LCL = 747.29 g). Root cause investigation confirmed a mid-shift fill-head adjustment was performed without following the change-control procedure. This confirmed special-cause variation — Subgroup 8 was legitimately removed and the analysis re-run on the 19-subgroup stable dataset.

Weight Distribution — Normality & Histogram



Anderson-Darling p-value = 0.838 ($p > 0.05$): data is normally distributed — Cp/Cpk indices are valid. The distribution is narrower than the specification window but visibly shifted below the 750 g target, explaining the Cpk shortfall.

Capability Indices vs. Requirement



Cp = 1.41 > 1.33 confirms sufficient process spread — the process fits within the specification window. Cpk = 1.24 < 1.33 confirms the process is OFF-CENTRE. The entire capability gap is a centering problem, not a spread problem.

ACTION PLAN

Recommendations

Immediate Actions (0–2 weeks)

- **Re-centre the fill head set-point by +0.56 g**: shifting the process mean from 749.44 g to 750.00 g will increase Cpk from 1.24 to ~1.41 — exceeding the 1.33 requirement with margin.
- **Enforce change-control procedure for fill-head adjustments**: the Subgroup 8 OC event was caused by an uncontrolled adjustment. Require supervisor sign-off and a 5-sample verification measurement before resuming production after any adjustment.
- **Implement Xbar-R chart for ongoing monitoring**: use UCL=751.47 g, LCL=747.29 g, CL=749.44 g (stable process). Define a written reaction plan: two consecutive points beyond 2-sigma = investigate; one point beyond UCL/LCL = stop and correct.

Short-Term Actions (2–8 weeks)

- **Reduce within-subgroup variation (target σ < 1.0 g)**: current within StDev = 1.182 g. Commission a fill-head maintenance inspection — worn dosing mechanism components are the most likely cause of within-shift variation.
- **Validate the weigh scale measurement system**: conduct a brief Gauge R&R; study to confirm measurement variation is not contributing to the 1.182 g within-subgroup StDev estimate.

Long-Term (2–6 months)

- **Target Cpk ≥ 1.50**: once centred and stabilised, continue variation-reduction work to build in margin against future drift — reducing σ from 1.182 g to 0.95 g achieves Cpk = 1.75 with no mean shift.
- **Extend to other fill lines**: apply the same Capability Sixpack methodology to all packaging lines and establish a company-wide Cpk tracking dashboard.

BUSINESS VALUE

Financial Impact & Expected Benefits

~88 PPM	<10 PPM	0.56 g	~$30K
Current Defect Rate	Post-Centering Target	Mean Shift Required	Est. Annual Savings

Category	Current State	Improved State	Benefit
Defect rate	~87.7 PPM within	<10 PPM	~89% fallout reduction
Customer complaints	4–6 cases/year	0–1 cases/year	Protect brand equity
Overfill waste	0.56 g/bag average	0 g/bag	Raw material savings
Regulatory risk	Cpk 1.24 — below customer minimum	Cpk 1.38+ — compliant	Audit pass
SPC monitoring	None in place	Xbar-R chart live	Real-time detection

- **Overfill cost:** at 0.56 g average overfill, a line producing 50,000 bags/day wastes 28 kg of rice per day — at $1.20/kg that is ~$12,000/year per line from mean shift alone.
- **Customer complaint handling:** each formal customer complaint in food packaging typically costs $5,000–15,000 in investigation, corrective action reports, and relationship management.
- **Regulatory exposure:** confirmed Cpk below customer minimum creates PPAP/re-approval risk — process re-qualification costs can reach $50,000–100,000 per line.

REFLECTION

Lessons Learned & Future Improvements

Lessons Learned

- **Stability must be verified before capability:** interpreting Cpk on an unstable process (including the OC subgroup) would have produced a misleading result. The Xbar-R chart step is non-negotiable.
- **The Cp vs. Cpk gap tells the story:** Cp = 1.41 and Cpk = 1.24 immediately communicated that the process spread was adequate — the entire problem was centering. This guided the recommendation directly to fill-head adjustment rather than expensive process-spread reduction work.
- **Normality testing matters:** without confirming normality (AD p=0.838), using Cp/Cpk would have been statistically unjustified. A non-normal distribution requires transformation or non-parametric capability methods.
- **Change control as a quality tool:** the OC event was not a process failure — it was a change-control failure. The process was capable of the target; the deviation came from an uncontrolled human intervention. Good SPC exposes both.

Future Improvements

- Conduct a Design of Experiments (DOE) on fill-head parameters to identify the optimal set-point and minimize sensitivity to environmental factors (ambient temperature, humidity affecting rice density).
- Implement automated weight feedback from the checkweigher into the SPC system to enable real-time Cpk trending without manual data entry.

Core Competencies Applied

Skill / Competency	Evidence in This Project
Process Capability Analysis	Cp, Cpk, Pp, Ppk computed and interpreted; correctly identified centering vs. spread issue
Statistical Process Control	Xbar-R chart; OC signal identified, root-caused, and legitimately removed
Normality Testing	Anderson-Darling test applied; assumption validated before using normal capability indices
Minitab Capability Sixpack	Full 6-panel capability analysis; Stat > Quality Tools > Capability Analysis

DMAIC Problem-Solving	End-to-end structured framework from complaint to quantified recommendation
Data-Driven Decision Making	Quantified the 0.56 g centering gap; financial impact estimated from PPM and waste data
Technical Communication	Statistical findings translated into business-language recommendations